higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PRINCIPLES OF COMPUTER PROGRAMMING

(First Paper)

NQF LEVEL 3

(10041023)

4 March 2019 (X-Paper)
09:00–11:00

This question paper consists of 9 pages.

NCV

Copyright reserved

TIME: 2 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Bistable multivibrators are also called …

A registers.
B astable.
C clocks.
D latches.

1.1.2 What will be the output of the following binary subtraction:

10001 – 1111

A 11000
B 10011
C 0010
D 1011

1.1.3 A/An … remains the same throughout the execution of a program.

A variable
B constant
C flag
D rapid application development

1.1.4 A … receives a parameter and returns a value.

A parameter
B subprocedure
C function
D logical expression

1.1.5 What will be the output of (7 < 2) AND (5 > 3)?

A True
B NOR
C Infinite
D False

Copyright reserved

1.1.6 An early program version with all the screen objects and labels, but without the event handlers at the stage

A RAD
B User Centered Design
C Prototype
D Desktop

1.1.7 A type of relationship in which a row in a table is linked to many rows in another table

A One to one
B One to many
C Many to one
D Many to many

1.1.8 The bubble sort method works through data by …

A swapping the smallest value with the first position in an array.
B deleting data in an array.
C repeatedly moving through the list of items until it is sorted.
D deleting the first and the last item.

1.1.9 Integer underflow errors are caused by …

A very big numbers.
B complicated variables.
C negative numbers.
D very small numbers.

1.1.10 As a programmer, you implement ... to guide the user to enter the data in the correct format.

A input masks
B validation rules
C induced tests
D range checks

(10 × 1) (10)

Copyright reserved

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 The output of the expression (5=2) OR (6=6) is false**.**

1.2.2 A global variable will be visible throughout the program.

1.2.3 An inquisitive user jumps ahead too fast without waiting for the program to finish its task.

1.2.4 Floating point errors occur when the exponent of the number has some missing digits.

1.2.5 A simple form allows you to manipulate data in more than one table at the same time.

1.2.6 A selection structure repeats a piece of code.

1.2.7 An array is an example of a text file.

1.2.8 A visual programming language (VPL) uses visual representation for what could have been done textually.

1.2.9 MS Word is an example of a database data structure.

1.2.10 A sequential structure accesses its data one after the other.

(10 × 1) (10)

1.3 Give one word or term for each of the following descriptions. Write only the word or term next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 A number system which has a base of sixteen

1.3.2 A type of error caused by an index that is bigger or smaller than the array dimension of the array reference in which it is used

1.3.3 A language that is used to query a database

1.3.4 A type of relationship that corresponds to only one record in another table

1.3.5 A characteristic that describes an entity, object or data structure in more detail

(5 × 1) (5)

[25]

TOTAL SECTION A: 25

Copyright reserved

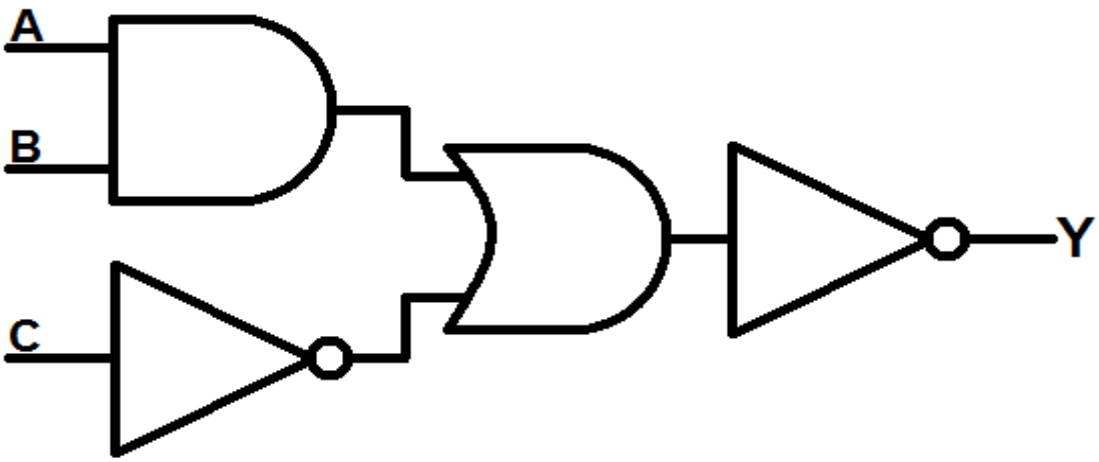
SECTION B

QUESTION 2

2.1 Find the 2's complement of the following:

11100111 (4)

2.2 Derive a logical expression from the following diagram:



(4)

2.3 Distinguish between a *function* and a *procedure*. (2)

2.4 Distinguish between a *variable* and a *constant*. (2 × 2) (4)

[14]

NCV

QUESTION 3

3.1 Study the following VB.Net code snippet and develop a complete trace table using the following heading:

Line Number	i	Mark(i)	Mark() >89	Mark() >79	Mark() >69	Mark() >59	Mark() >49	Mark() >39	Other	Symbol	Output

```
1  Public Class Form1
2      Dim mark() As Integer = {20, 90, 45, 66, 2, 55}
3      Dim symbol As Char
4      Private Sub btnDisplay_Click(sender As System.Object, e As System.
       Click
5          For i As Integer = 0 To mark.Length - 1
6              lstDisplay.Items.Add(mark(i).ToString.PadRight(5) & symbol
7              If mark(i) > 89 Then
8                  symbol = "A"
9
10             ElseIf mark(i) > 79 Then
11                 symbol = "B"
12
13             ElseIf mark(i) > 69 Then
14                 symbol = "C"
15             ElseIf mark(i) > 59 Then
16                 symbol = "D"
17             ElseIf mark(i) > 49 Then
18                 symbol = "E"
19
20             ElseIf mark(i) > 39 Then
21                 symbol = "F"
22             Else
23                 symbol = "G"
24             End If
25         Next
26     End Sub
27 End Class
```

(9)

3.2 Name TWO causes of conversion errors. (2)

3.3 Define *word size* in computer architecture. (2)

[13]

Copyright reserved

QUESTION 4

4.1 Study the following database table and answer the questions.

OurInfo				
ID	**FullName**	**Age**	**Gender**	**Province**
1	Masakala L	28	M	Limpopo
2	Lotz R	27	M	Limpopo
3	Makoe MR	38	M	Gauteng
4	Cina MM	29	F	Gauteng
5	Radebe B	30	F	Mpumalanga
6	Moeketsi NS	27	M	Gauteng
7	Sephooa M	25	M	Gauteng

4.1.1 Write an SQL statement that will extract all the males from Gauteng below the age of 40. (5)

4.1.2 Which field can be used as a primary key and why? (2 × 1) (2)

4.2 Distinguish between *primary key*, *alternate key* and *foreign key*. (3 × 2) (6)

[13]

QUESTION 5

Study the following piece of code and answer the questions.

```
Dim number as integer = 2
For i As Integer = 1 To 5
        lstDisplay.Items.Add(i & "x" & number & "=" & number * i)
    Next
```

5.1 What will be the output on the listbox after the program is run? Show ALL outputs. (5)

5.2 List THREE programming structures. (3)

5.3 List THREE program requirements for an IPO chart. (3)

[11]

QUESTION 6

6.1 Explain what is meant by *defensive programming.* (3)

6.2 Explain how you will deal with each of the following types of users when applying defensive programming in your program:

6.2.1 Inquisitive

6.2.2 Impatient

6.2.3 Thoughtless

6.2.4 Scared and nervous

6.2.5 Expert and skilled

(5 × 2) (10)
[13]

QUESTION 7

7.1 Give ONE example of each of the following:

7.1.1 A logical data variable (1)

7.1.2 A signed floating number (2)

7.2 Give TWO features of each of the following data structures:

7.2.1 Text file

7.2.2 Array

7.2.3 Database

(3 × 2) (6)

7.3 Explain what is meant by *abstract data type*. (2)
[11]

TOTAL SECTION B: 75
GRAND TOTAL: 100

Copyright reserved